Md. Mosharaf Hossain Apurbo

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Research Interests

NeuroAI, brain-inspired and neuromorphic learning, graph representation learning, world models, self-improving AI, robust and efficient computer vision, adversarial robustness of vision models, and on-device ML.

Education

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering (Level-4) *Nov 2022 – Present*
Advisor: Dr. Atif Hasan Rahman (Associate Professor, Dept. of CSE, BUET)

Notre Dame College Dhaka, Bangladesh
Higher Secondary Certificate (HSC), GPA 5.00/5.00 *Jul 2019 – Dec 2021*

Research Experience

Prescribed Degree Distributions in Chordal Graphs of Bounded Clique Number Ongoing
Undergraduate Thesis, BUET — Supervisor: Dr. Md. Saidur Rahman (Professor, Dept. of CSE) *Jun 2026 – Present*

- Exactly characterised which degree distributions are achievable by chordal graphs of bounded clique number, and showed the same condition holds for k-degenerate, bounded-treewidth and bounded-pathwidth graphs.
- Designed a **linear-time generator** that outputs chordal graphs with a prescribed degree distribution (with a perfect elimination order as certificate); for power laws it attains the **optimal convergence rate** with no degree cut-off.

Research Collaboration with Samsung R&D Institute Bangladesh Dhaka, Bangladesh
Industry-sponsored CSE-450 Capstone Project, BUET — Smart-glasses AI *Dec 2025 – Sep 2026*

- Developed a lightweight on-device model for pointing-based visual grounding, deployed on Samsung's WearOS-based smart glasses.
- Resulted in a paper under review at ACM IMWUT (see Publications) and 2nd Runners-up at Samsung's AI Proof of Concept Exhibition.

Publications

1. Md. Mosharaf Hossain Apurbo, Md. Asif Ali, and Kazi Reyazul Hasan. **Multiple-Instance Contrastive Alignment for Pose-Based Sign Spotting**. In *Proceedings of the Second Workshop on Sign Language Processing (WSLP 2026)*, co-located with EMNLP 2026, Budapest, Hungary. [Paper] Published
2. Md. Asif Ali, Md. Mosharaf Hossain Apurbo, and Kazi Reyazul Hasan. **Metric-Learning Retrieval for Cross-Signer Isolated Sign Language Recognition**. In *Proceedings of the Second Workshop on Sign Language Processing (WSLP 2026)*, co-located with EMNLP 2026, Budapest, Hungary. [Paper] Published
3. Md. Mosharaf Hossain Apurbo*, Niloy Das Robin*, Nabiha Tahseen, Gourove Roy, Sayjad Rahman, Farhan Tanvir, Avik Saha, Mohammad Abrar Nafee Akhand, and Sadia Sharmin. **DeicNet: On-Device Pointing-Based Visual Grounding for Smart Glasses**. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2026. Under Review
4. Md. Mahedi Hasan Rigan, Md. Mosharaf Hossain Apurbo, Md. Olid Hasan Bhuiyan, and Towhidul Islam. **Which Object Is Fake? Removing Shortcuts from Object-Level AI-Generated Image Detection**. Submitted to the *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2027. Under Review

*Equal contribution.

Selected Projects

Trip.AI — AI Travel Planning Platform Oct 2024
Agentic AI application

- Built **agentic, personalised itinerary generation** that plans trips around each user's preferences, with real-time weather updates.
- Implemented **RAG-based natural-language image search** during the BUET CSE Fest Hackathon 2024 (**5th place** and Best Documentation award).
- Added AI-generated travel blogs and vlogs with accompanying music.

Mini-PC: 8-bit MIPS Processor (Logisim) Oct 2024
Computer architecture

- Designed an 8-bit MIPS processor from scratch: control logic, ALU, memory access and I/O for a complete instruction cycle.
- Built a display module and created a **Snake game** that runs on the processor.

C-to-Intel-8086 Compiler (C++17, ANTLR 4) Jan 2025 – Jun 2025
Low-level systems programming

- Full front end (lexer, semantic analysis, scoped symbol table) and 8086 back end with BP-relative stack frames, recursion and arrays; peephole optimiser.

Quantum Machine Learning Benchmarking for Biomedical Data
Scientific ML

- Benchmarking framework for quantum feature encodings and ansatz models with SHAP/LASSO/PCA feature selection and convergence visualisations.

BookSphere — E-commerce Platform Jan 2024 – Feb 2024
Full-stack web application

- An Oracle-powered, Bootstrap-based e-commerce site replicating Rokomari.com's core features.

Honours

- 2nd Runners-up, AI Proof of Concept Exhibition, Samsung R&D Institute Bangladesh, Sep 2026.
- 5th Place and Best Documentation (AI/API), BUET CSE Fest Hackathon 2024.
- Merit Talent Scholarship, Secondary School Certificate (2019) and Junior Secondary School Certificate (2017), Dhaka Board.

Technical Skills

Programming Languages	Python, JavaScript, C#, Java, C++, Assembly
Deep Learning Frameworks	PyTorch, TensorFlow, Keras, CUDA libraries
Machine Learning Frameworks	Scikit-learn, XGBoost, LightGBM, SciPy
Libraries	Hugging Face Transformers, OpenCV, NumPy, Pandas, Matplotlib, Seaborn
Web Development	React, Bootstrap, Django, REST, HTML, CSS, Node.js
Developer Tools & Platforms	Git, Docker, Kubernetes, .NET, UWP
Database Management	Cassandra, MySQL, SQLite, Oracle